Linux Text Processing Task: grep, cut, awk

Team Name: Whale_Riders

Attendance: 5

Sample log (access.log)

The log is a web server access log. Each line holds: client IP, timestamp, request (method, path, protocol), status code, response size in bytes, and user agent.

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203.0.113.45 - - [05/Nov/2026:10:01:01 +0530] "GET /missing.html HTTP/1.1" 404 243
"python-requests/2.31"
192.168.1.10 - - [05/Nov/2026:10:01:48 +0530] "GET /api/orders HTTP/1.1" 500 330
"Mozilla/5.0"
10.0.0.5 - - [05/Nov/2026:10:02:36 +0530] "GET /about.html HTTP/1.1" 200 10255
"Mozilla/5.0"
192.168.1.10 - - [05/Nov/2026:10:03:49 +0530] "POST /api/login HTTP/1.1" 200 10153
"Googlebot/2.1"
192.168.1.15 - - [05/Nov/2026:10:04:33 +0530] "GET /images/logo.png HTTP/1.1" 200
20573 "curl/8.4.0"
192.168.1.15 - - [05/Nov/2026:10:05:51 +0530] "PUT /api/users/12 HTTP/1.1" 200 13815
"Mozilla/5.0"
192.168.1.15 - - [05/Nov/2026:10:07:08 +0530] "GET /images/logo.png HTTP/1.1" 200
21454 "Mozilla/5.0"
10.0.0.8 - - [05/Nov/2026:10:07:37 +0530] "GET /Admin/settings HTTP/1.1" 403 327
"Mozilla/5.0"
192.168.1.22 - - [05/Nov/2026:10:08:31 +0530] "GET /images/logo.png HTTP/1.1" 200
21160 "Googlebot/2.1"
192.168.1.10 - - [05/Nov/2026:10:09:35 +0530] "POST /api/orders HTTP/1.1" 503 447
"Mozilla/5.0"
192.168.1.15 - - [05/Nov/2026:10:10:26 +0530] "GET /contact.html HTTP/1.1" 200 2120
"python-requests/2.31"
10.0.0.5 - - [05/Nov/2026:10:11:21 +0530] "GET /old-page HTTP/1.1" 301 209
"Googlebot/2.1"
192.168.1.10 - - [05/Nov/2026:10:12:10 +0530] "GET /api/users HTTP/1.1" 200 5104
"python-requests/2.31"
203.0.113.45 - - [05/Nov/2026:10:13:10 +0530] "GET /favicon.ico HTTP/1.1" 404 293
"curl/8.4.0"
192.168.1.10 - - [05/Nov/2026:10:14:20 +0530] "GET /missing.html HTTP/1.1" 404 191
"python-requests/2.31"
192.168.1.10 - - [05/Nov/2026:10:14:46 +0530] "POST /api/login HTTP/1.1" 401 257
"curl/8.4.0"
10.0.0.5 - - [05/Nov/2026:10:15:25 +0530] "GET /old-page HTTP/1.1" 301 271
"curl/8.4.0"
192.168.1.10 - - [05/Nov/2026:10:16:24 +0530] "POST /api/login HTTP/1.1" 200 13082
"Mozilla/5.0"
192.168.1.15 - - [05/Nov/2026:10:17:51 +0530] "GET /index.html HTTP/1.1" 200 13221
"Mozilla/5.0"
10.0.0.8 - - [05/Nov/2026:10:18:56 +0530] "GET /favicon.ico HTTP/1.1" 404 199
"Googlebot/2.1"
```

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192.168.1.15 - - [05/Nov/2026:10:19:45 +0530] "GET /contact.html HTTP/1.1" 200 6625
"Googlebot/2.1"
192.168.1.22 - - [05/Nov/2026:10:20:49 +0530] "GET /old-page HTTP/1.1" 301 206
"Mozilla/5.0"
10.0.0.5 - - [05/Nov/2026:10:22:10 +0530] "GET /index.html HTTP/1.1" 200 8655
"Googlebot/2.1"
192.168.1.22 - - [05/Nov/2026:10:22:55 +0530] "GET /api/users HTTP/1.1" 200 7909
"Googlebot/2.1"
10.0.0.8 - - [05/Nov/2026:10:23:25 +0530] "GET /api/users HTTP/1.1" 200 2881
"curl/8.4.0"
192.168.1.10 - - [05/Nov/2026:10:24:29 +0530] "POST /api/login HTTP/1.1" 401 234
"Mozilla/5.0"
192.168.1.15 - - [05/Nov/2026:10:24:52 +0530] "GET /admin HTTP/1.1" 403 266
"Mozilla/5.0"
172.16.0.3 - - [05/Nov/2026:10:25:57 +0530] "GET /favicon.ico HTTP/1.1" 404 284
"curl/8.4.0"
10.0.0.5 - - [05/Nov/2026:10:26:40 +0530] "POST /api/login HTTP/1.1" 401 258
"Mozilla/5.0"
192.168.1.22 - - [05/Nov/2026:10:28:06 +0530] "GET /contact.html HTTP/1.1" 200 13649
"Mozilla/5.0"
192.168.1.22 - - [05/Nov/2026:10:28:38 +0530] "GET /index.html HTTP/1.1" 200 13251
"Mozilla/5.0"
192.168.1.22 - - [05/Nov/2026:10:29:33 +0530] "GET /contact.html HTTP/1.1" 200 9119
"curl/8.4.0"
192.168.1.15 - - [05/Nov/2026:10:30:08 +0530] "GET /Admin/settings HTTP/1.1" 403 218
"Mozilla/5.0"
203.0.113.45 - - [05/Nov/2026:10:30:47 +0530] "GET /old-page HTTP/1.1" 301 185
"Mozilla/5.0"
192.168.1.10 - - [05/Nov/2026:10:31:35 +0530] "GET /api/orders HTTP/1.1" 500 463
"Googlebot/2.1"
192.168.1.10 - - [05/Nov/2026:10:32:41 +0530] "POST /api/orders HTTP/1.1" 503 451
"python-requests/2.31"
192.168.1.10 - - [05/Nov/2026:10:33:09 +0530] "PUT /api/users/12 HTTP/1.1" 200 2516
"Mozilla/5.0"
172.16.0.3 - - [05/Nov/2026:10:34:23 +0530] "DELETE /api/users/7 HTTP/1.1" 204 0
"Mozilla/5.0"
192.168.1.10 - - [05/Nov/2026:10:35:18 +0530] "GET /api/users HTTP/1.1" 200 7768
"python-requests/2.31"
192.168.1.10 - - [05/Nov/2026:10:36:06 +0530] "POST /api/login HTTP/1.1" 200 8234
"Mozilla/5.0"
10.0.0.5 - - [05/Nov/2026:10:37:33 +0530] "POST /api/login HTTP/1.1" 200 3445
"python-requests/2.31"
192.168.1.15 - - [05/Nov/2026:10:38:19 +0530] "GET /favicon.ico HTTP/1.1" 404 214
"Mozilla/5.0"
10.0.0.5 - - [05/Nov/2026:10:39:43 +0530] "POST /api/users HTTP/1.1" 201 443
"python-requests/2.31"
203.0.113.45 - - [05/Nov/2026:10:40:53 +0530] "GET /admin HTTP/1.1" 403 250
"Mozilla/5.0"
192.168.1.10 - - [05/Nov/2026:10:41:29 +0530] "DELETE /api/users/7 HTTP/1.1" 204 0
"Mozilla/5.0"
10.0.0.5 - - [05/Nov/2026:10:42:20 +0530] "GET /about.html HTTP/1.1" 200 8327
"Mozilla/5.0"
192.168.1.15 - - [05/Nov/2026:10:43:50 +0530] "GET /about.html HTTP/1.1" 200 5871
"curl/8.4.0"
192.168.1.22 - - [05/Nov/2026:10:44:45 +0530] "POST /api/users HTTP/1.1" 201 458
```

```
"python-requests/2.31"
192.168.1.15 - - [05/Nov/2026:10:45:10 +0530] "GET /admin HTTP/1.1" 403 349
"Googlebot/2.1"
10.0.0.8 - - [05/Nov/2026:10:46:19 +0530] "GET /contact.html HTTP/1.1" 200 3248
"Googlebot/2.1"
```

Section A: grep (Questions 1 to 10)

Q1. Show every request that returned status code 404.

Command: grep '404' access.log

Output: 203.0.113.45 - - [05/Nov/2026:10:01:01 +0530] "GET /missing.html HTTP/1.1" 404 243 "python-requests/2.31"

203.0.113.45 - - [05/Nov/2026:10:13:10 +0530] "GET /favicon.ico HTTP/1.1" 404 293 "curl/8.4.0"

192.168.1.10 - - [05/Nov/2026:10:14:20 +0530] "GET /missing.html HTTP/1.1" 404 191 "python-requests/2.31"

10.0.0.8 - - [05/Nov/2026:10:18:56 +0530] "GET /favicon.ico HTTP/1.1" 404 199 "Googlebot/2.1"

172.16.0.3 - - [05/Nov/2026:10:25:57 +0530] "GET /favicon.ico HTTP/1.1" 404 284 "curl/8.4.0"

192.168.1.15 - - [05/Nov/2026:10:38:19 +0530] "GET /favicon.ico HTTP/1.1" 404 214 "Mozilla/5.0"

203.0.113.45 - - [05/Nov/2026:10:01:01 +0530] "GET /missing.html HTTP/1.1" 404 243 "python-requests/2.31"

203.0.113.45 - - [05/Nov/2026:10:13:10 +0530] "GET /favicon.ico HTTP/1.1" 404 293 "curl/8.4.0"

192.168.1.10 - - [05/Nov/2026:10:14:20 +0530] "GET /missing.html HTTP/1.1" 404 191 "python-requests/2.31"

10.0.0.8 - - [05/Nov/2026:10:18:56 +0530] "GET /favicon.ico HTTP/1.1" 404 199 "Googlebot/2.1"

172.16.0.3 - - [05/Nov/2026:10:25:57 +0530] "GET /favicon.ico HTTP/1.1" 404 284 "curl/8.4.0"

192.168.1.15 - - [05/Nov/2026:10:38:19 +0530] "GET /favicon.ico HTTP/1.1" 404 214 "Mozilla/5.0"

Q2. Count how many POST requests are in the log.

Command: grep -c 'POST' access.log

Output: 22

Q3. Show all requests made from the IP address 192.168.1.15.

Command: grep '^192\.168\.1\.15' access.log

Output:192.168.1.15 - - [05/Nov/2026:10:04:33 +0530] "GET /images/logo.png HTTP/1.1" 200 20573 "curl/8.4.0"

192.168.1.15 - - [05/Nov/2026:10:05:51 +0530] "PUT /api/users/12 HTTP/1.1" 200 13815 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:07:08 +0530] "GET /images/logo.png HTTP/1.1" 200 21454 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:10:26 +0530] "GET /contact.html HTTP/1.1" 200 2120 "python-requests/2.31"

192.168.1.15 - - [05/Nov/2026:10:17:51 +0530] "GET /index.html HTTP/1.1" 200 13221 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:19:45 +0530] "GET /contact.html HTTP/1.1" 200 6625 "Googlebot/2.1"

192.168.1.15 - - [05/Nov/2026:10:24:52 +0530] "GET /admin HTTP/1.1" 403 266 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:30:08 +0530] "GET /Admin/settings HTTP/1.1" 403 218 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:38:19 +0530] "GET /favicon.ico HTTP/1.1" 404 214 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:43:50 +0530] "GET /about.html HTTP/1.1" 200 5871 "curl/8.4.0"

192.168.1.15 - - [05/Nov/2026:10:45:10 +0530] "GET /admin HTTP/1.1" 403 349 "Googlebot/2.1"

192.168.1.15 - - [05/Nov/2026:10:04:33 +0530] "GET /images/logo.png HTTP/1.1" 200 20573 "curl/8.4.0"

192.168.1.15 - - [05/Nov/2026:10:05:51 +0530] "PUT /api/users/12 HTTP/1.1" 200 13815 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:07:08 +0530] "GET /images/logo.png HTTP/1.1" 200 21454 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:10:26 +0530] "GET /contact.html HTTP/1.1" 200 2120 "python-requests/2.31"

192.168.1.15 - - [05/Nov/2026:10:17:51 +0530] "GET /index.html HTTP/1.1" 200 13221 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:19:45 +0530] "GET /contact.html HTTP/1.1" 200 6625 "Googlebot/2.1"

192.168.1.15 - - [05/Nov/2026:10:24:52 +0530] "GET /admin HTTP/1.1" 403 266 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:30:08 +0530] "GET /Admin/settings HTTP/1.1" 403 218 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:38:19 +0530] "GET /favicon.ico HTTP/1.1" 404 214 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:43:50 +0530] "GET /about.html HTTP/1.1" 200 5871 "curl/8.4.0"

192.168.1.15 - - [05/Nov/2026:10:45:10 +0530] "GET /admin HTTP/1.1" 403 349 "Googlebot/2.1"

Q4. Show every line that does NOT have status code 200.

Command: grep -v '200' access.log

Output: 203.0.113.45 - - [05/Nov/2026:10:01:01 +0530] "GET /missing.html HTTP/1.1" 404 243 "python-requests/2.31"

192.168.1.10 - - [05/Nov/2026:10:01:48 +0530] "GET /api/orders HTTP/1.1" 500 330 "Mozilla/5.0"

10.0.0.8 - - [05/Nov/2026:10:07:37 +0530] "GET /Admin/settings HTTP/1.1" 403 327 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:09:35 +0530] "POST /api/orders HTTP/1.1" 503 447 "Mozilla/5.0"

10.0.0.5 - - [05/Nov/2026:10:11:21 +0530] "GET /old-page HTTP/1.1" 301 209 "Googlebot/2.1"

203.0.113.45 - - [05/Nov/2026:10:13:10 +0530] "GET /favicon.ico HTTP/1.1" 404 293 "curl/8.4.0"

192.168.1.10 - - [05/Nov/2026:10:14:20 +0530] "GET /missing.html HTTP/1.1" 404 191 "python-requests/2.31"

192.168.1.10 - - [05/Nov/2026:10:14:46 +0530] "POST /api/login HTTP/1.1" 401 257 "curl/8.4.0"

10.0.0.5 - - [05/Nov/2026:10:15:25 +0530] "GET /old-page HTTP/1.1" 301 271 "curl/8.4.0"

10.0.0.8 - - [05/Nov/2026:10:18:56 +0530] "GET /favicon.ico HTTP/1.1" 404 199 "Googlebot/2.1"

192.168.1.22 - - [05/Nov/2026:10:20:49 +0530] "GET /old-page HTTP/1.1" 301 206 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:24:29 +0530] "POST /api/login HTTP/1.1" 401 234 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:24:52 +0530] "GET /admin HTTP/1.1" 403 266 "Mozilla/5.0"

172.16.0.3 - - [05/Nov/2026:10:25:57 +0530] "GET /favicon.ico HTTP/1.1" 404 284 "curl/8.4.0"

10.0.0.5 - - [05/Nov/2026:10:26:40 +0530] "POST /api/login HTTP/1.1" 401 258 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:30:08 +0530] "GET /Admin/settings HTTP/1.1" 403 218 "Mozilla/5.0"

203.0.113.45 - - [05/Nov/2026:10:30:47 +0530] "GET /old-page HTTP/1.1" 301 185 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:31:35 +0530] "GET /api/orders HTTP/1.1" 500 463 "Googlebot/2.1"

192.168.1.10 - - [05/Nov/2026:10:32:41 +0530] "POST /api/orders HTTP/1.1" 503 451 "python-requests/2.31"

172.16.0.3 - - [05/Nov/2026:10:34:23 +0530] "DELETE /api/users/7 HTTP/1.1" 204 0 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:38:19 +0530] "GET /favicon.ico HTTP/1.1" 404 214 "Mozilla/5.0"

10.0.0.5 - - [05/Nov/2026:10:39:43 +0530] "POST /api/users HTTP/1.1" 201 443 "python-requests/2.31"

203.0.113.45 - - [05/Nov/2026:10:40:53 +0530] "GET /admin HTTP/1.1" 403 250 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:41:29 +0530] "DELETE /api/users/7 HTTP/1.1" 204 0 "Mozilla/5.0"

192.168.1.22 - - [05/Nov/2026:10:44:45 +0530] "POST /api/users HTTP/1.1" 201 458 "python-requests/2.31"

192.168.1.15 - - [05/Nov/2026:10:45:10 +0530] "GET /admin HTTP/1.1" 403 349 "Googlebot/2.1"

203.0.113.45 - - [05/Nov/2026:10:01:01 +0530] "GET /missing.html HTTP/1.1" 404 243 "python-requests/2.31"

192.168.1.10 - - [05/Nov/2026:10:01:48 +0530] "GET /api/orders HTTP/1.1" 500 330 "Mozilla/5.0"

10.0.0.8 - - [05/Nov/2026:10:07:37 +0530] "GET /Admin/settings HTTP/1.1" 403 327 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:09:35 +0530] "POST /api/orders HTTP/1.1" 503 447 "Mozilla/5.0"

10.0.0.5 - - [05/Nov/2026:10:11:21 +0530] "GET /old-page HTTP/1.1" 301 209 "Googlebot/2.1"

203.0.113.45 - - [05/Nov/2026:10:13:10 +0530] "GET /favicon.ico HTTP/1.1" 404 293 "curl/8.4.0"

192.168.1.10 - - [05/Nov/2026:10:14:20 +0530] "GET /missing.html HTTP/1.1" 404 191 "python-requests/2.31"

192.168.1.10 - - [05/Nov/2026:10:14:46 +0530] "POST /api/login HTTP/1.1" 401 257 "curl/8.4.0"

10.0.0.5 - - [05/Nov/2026:10:15:25 +0530] "GET /old-page HTTP/1.1" 301 271 "curl/8.4.0"

10.0.0.8 - - [05/Nov/2026:10:18:56 +0530] "GET /favicon.ico HTTP/1.1" 404 199 "Googlebot/2.1"

192.168.1.22 - - [05/Nov/2026:10:20:49 +0530] "GET /old-page HTTP/1.1" 301 206 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:24:29 +0530] "POST /api/login HTTP/1.1" 401 234 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:24:52 +0530] "GET /admin HTTP/1.1" 403 266 "Mozilla/5.0"

172.16.0.3 - - [05/Nov/2026:10:25:57 +0530] "GET /favicon.ico HTTP/1.1" 404 284 "curl/8.4.0"

10.0.0.5 - - [05/Nov/2026:10:26:40 +0530] "POST /api/login HTTP/1.1" 401 258 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:30:08 +0530] "GET /Admin/settings HTTP/1.1" 403 218 "Mozilla/5.0"

203.0.113.45 - - [05/Nov/2026:10:30:47 +0530] "GET /old-page HTTP/1.1" 301 185 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:31:35 +0530] "GET /api/orders HTTP/1.1" 500 463 "Googlebot/2.1"

192.168.1.10 - - [05/Nov/2026:10:32:41 +0530] "POST /api/orders HTTP/1.1" 503 451 "python-requests/2.31"

172.16.0.3 - - [05/Nov/2026:10:34:23 +0530] "DELETE /api/users/7 HTTP/1.1" 204 0 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:38:19 +0530] "GET /favicon.ico HTTP/1.1" 404 214 "Mozilla/5.0"

10.0.0.5 - - [05/Nov/2026:10:39:43 +0530] "POST /api/users HTTP/1.1" 201 443 "python-requests/2.31"

203.0.113.45 - - [05/Nov/2026:10:40:53 +0530] "GET /admin HTTP/1.1" 403 250 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:41:29 +0530] "DELETE /api/users/7 HTTP/1.1" 204 0 "Mozilla/5.0"

192.168.1.22 - - [05/Nov/2026:10:44:45 +0530] "POST /api/users HTTP/1.1" 201 458 "python-requests/2.31"

192.168.1.15 - - [05/Nov/2026:10:45:10 +0530] "GET /admin HTTP/1.1" 403 349 "Googlebot/2.1"

Q5. Find every request to an admin page, whether it is written as "admin" or "Admin" (case-insensitive search).

Command:grep -i 'admin' access.log

Output:10.0.0.8 - - [05/Nov/2026:10:07:37 +0530] "GET /Admin/settings HTTP/1.1" 403 327 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:24:52 +0530] "GET /admin HTTP/1.1" 403 266 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:30:08 +0530] "GET /Admin/settings HTTP/1.1" 403 218 "Mozilla/5.0"

203.0.113.45 - - [05/Nov/2026:10:40:53 +0530] "GET /admin HTTP/1.1" 403 250 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:45:10 +0530] "GET /admin HTTP/1.1" 403 349 "Googlebot/2.1"

10.0.0.8 - - [05/Nov/2026:10:07:37 +0530] "GET /Admin/settings HTTP/1.1" 403 327 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:24:52 +0530] "GET /admin HTTP/1.1" 403 266 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:30:08 +0530] "GET /Admin/settings HTTP/1.1" 403 218 "Mozilla/5.0"

203.0.113.45 - - [05/Nov/2026:10:40:53 +0530] "GET /admin HTTP/1.1" 403 250 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:45:10 +0530] "GET /admin HTTP/1.1" 403 349 "Googlebot/2.1"

Q6. Show the line numbers of all requests that returned status 500.

Command:grep -n '500' access.log

Output:

2:192.168.1.10 - - [05/Nov/2026:10:01:48 +0530] "GET /api/orders HTTP/1.1" 500 330 "Mozilla/5.0"

35:192.168.1.10 - - [05/Nov/2026:10:31:35 +0530] "GET /api/orders HTTP/1.1" 500 463 "Googlebot/2.1"

Q7. Show all requests made using the curl user agent.

Command: grep 'curl/' access.log

Output:

192.168.1.15 - - [05/Nov/2026:10:04:33 +0530] "GET /images/logo.png HTTP/1.1" 200 20573 "curl/8.4.0"

203.0.113.45 - - [05/Nov/2026:10:13:10 +0530] "GET /favicon.ico HTTP/1.1" 404 293 "curl/8.4.0"

192.168.1.10 - - [05/Nov/2026:10:14:46 +0530] "POST /api/login HTTP/1.1" 401 257 "curl/8.4.0"

10.0.0.5 - - [05/Nov/2026:10:15:25 +0530] "GET /old-page HTTP/1.1" 301 271 "curl/8.4.0"

10.0.0.8 - - [05/Nov/2026:10:23:25 +0530] "GET /api/users HTTP/1.1" 200 2881 "curl/8.4.0"

172.16.0.3 - - [05/Nov/2026:10:25:57 +0530] "GET /favicon.ico HTTP/1.1" 404 284 "curl/8.4.0"

192.168.1.22 - - [05/Nov/2026:10:29:33 +0530] "GET /contact.html HTTP/1.1" 200 9119 "curl/8.4.0"

192.168.1.15 - - [05/Nov/2026:10:43:50 +0530] "GET /about.html HTTP/1.1" 200 5871 "curl/8.4.0"

Q8. Show all requests to API endpoints (paths starting with /api/).

Command: grep '/api/' access.log

Output:

192.168.1.10 - - [05/Nov/2026:10:01:48 +0530] "GET /api/orders HTTP/1.1" 500 330 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:03:49 +0530] "POST /api/login HTTP/1.1" 200 10153 "Googlebot/2.1"

192.168.1.15 - - [05/Nov/2026:10:05:51 +0530] "PUT /api/users/12 HTTP/1.1" 200 13815 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:09:35 +0530] "POST /api/orders HTTP/1.1" 503 447 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:12:10 +0530] "GET /api/users HTTP/1.1" 200 5104 "python-requests/2.31"

192.168.1.10 - - [05/Nov/2026:10:14:46 +0530] "POST /api/login HTTP/1.1" 401 257 "curl/8.4.0"

192.168.1.10 - - [05/Nov/2026:10:16:24 +0530] "POST /api/login HTTP/1.1" 200 13082 "Mozilla/5.0"

192.168.1.22 - - [05/Nov/2026:10:22:55 +0530] "GET /api/users HTTP/1.1" 200 7909 "Googlebot/2.1"

10.0.0.8 - - [05/Nov/2026:10:23:25 +0530] "GET /api/users HTTP/1.1" 200 2881 "curl/8.4.0"

192.168.1.10 - - [05/Nov/2026:10:24:29 +0530] "POST /api/login HTTP/1.1" 401 234 "Mozilla/5.0"

10.0.0.5 - - [05/Nov/2026:10:26:40 +0530] "POST /api/login HTTP/1.1" 401 258 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:31:35 +0530] "GET /api/orders HTTP/1.1" 500 463 "Googlebot/2.1"

192.168.1.10 - - [05/Nov/2026:10:32:41 +0530] "POST /api/orders HTTP/1.1" 503 451 "python-requests/2.31"

192.168.1.10 - - [05/Nov/2026:10:33:09 +0530] "PUT /api/users/12 HTTP/1.1" 200 2516 "Mozilla/5.0"

172.16.0.3 - - [05/Nov/2026:10:34:23 +0530] "DELETE /api/users/7 HTTP/1.1" 204 0 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:35:18 +0530] "GET /api/users HTTP/1.1" 200 7768 "python-requests/2.31"

192.168.1.10 - - [05/Nov/2026:10:36:06 +0530] "POST /api/login HTTP/1.1" 200 8234 "Mozilla/5.0"

10.0.0.5 - - [05/Nov/2026:10:37:33 +0530] "POST /api/login HTTP/1.1" 200 3445 "python-requests/2.31"

10.0.0.5 - - [05/Nov/2026:10:39:43 +0530] "POST /api/users HTTP/1.1" 201 443 "python-requests/2.31"

192.168.1.10 - - [05/Nov/2026:10:41:29 +0530] "DELETE /api/users/7 HTTP/1.1" 204 0 "Mozilla/5.0"

192.168.1.22 - - [05/Nov/2026:10:44:45 +0530] "POST /api/users HTTP/1.1" 201 458 "python-requests/2.31"

Q9. Count all error responses, meaning any 4xx or 5xx status code. Hint: use an extended regular expression.

Command: grep -E '[45][0-9][0-9]' access.log

Output:

203.0.113.45 - - [05/Nov/2026:10:01:01 +0530] "GET /missing.html HTTP/1.1" 404 243 "python-requests/2.31"

192.168.1.10 - - [05/Nov/2026:10:01:48 +0530] "GET /api/orders HTTP/1.1" 500 330 "Mozilla/5.0"

10.0.0.5 - - [05/Nov/2026:10:02:36 +0530] "GET /about.html HTTP/1.1" 200 10255 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:03:49 +0530] "POST /api/login HTTP/1.1" 200 10153 "Googlebot/2.1"

192.168.1.15 - - [05/Nov/2026:10:04:33 +0530] "GET /images/logo.png HTTP/1.1" 200 20573 "curl/8.4.0"

192.168.1.15 - - [05/Nov/2026:10:05:51 +0530] "PUT /api/users/12 HTTP/1.1" 200 13815 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:07:08 +0530] "GET /images/logo.png HTTP/1.1" 200 21454 "Mozilla/5.0"

10.0.0.8 - - [05/Nov/2026:10:07:37 +0530] "GET /Admin/settings HTTP/1.1" 403 327 "Mozilla/5.0"

192.168.1.22 - - [05/Nov/2026:10:08:31 +0530] "GET /images/logo.png HTTP/1.1" 200 21160 "Googlebot/2.1"

192.168.1.10 - - [05/Nov/2026:10:09:35 +0530] "POST /api/orders HTTP/1.1" 503 447 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:10:26 +0530] "GET /contact.html HTTP/1.1" 200 2120 "python-requests/2.31"

10.0.0.5 - - [05/Nov/2026:10:11:21 +0530] "GET /old-page HTTP/1.1" 301 209 "Googlebot/2.1"

192.168.1.10 - - [05/Nov/2026:10:12:10 +0530] "GET /api/users HTTP/1.1" 200 5104 "python-requests/2.31"

203.0.113.45 - - [05/Nov/2026:10:13:10 +0530] "GET /favicon.ico HTTP/1.1" 404 293 "curl/8.4.0"

192.168.1.10 - - [05/Nov/2026:10:14:20 +0530] "GET /missing.html HTTP/1.1" 404 191 "python-requests/2.31"

192.168.1.10 - - [05/Nov/2026:10:14:46 +0530] "POST /api/login HTTP/1.1" 401 257 "curl/8.4.0"

10.0.0.5 - - [05/Nov/2026:10:15:25 +0530] "GET /old-page HTTP/1.1" 301 271 "curl/8.4.0"

192.168.1.10 - - [05/Nov/2026:10:16:24 +0530] "POST /api/login HTTP/1.1" 200 13082 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:17:51 +0530] "GET /index.html HTTP/1.1" 200 13221 "Mozilla/5.0"

10.0.0.8 - - [05/Nov/2026:10:18:56 +0530] "GET /favicon.ico HTTP/1.1" 404 199 "Googlebot/2.1"

192.168.1.15 - - [05/Nov/2026:10:19:45 +0530] "GET /contact.html HTTP/1.1" 200 6625 "Googlebot/2.1"

192.168.1.22 - - [05/Nov/2026:10:20:49 +0530] "GET /old-page HTTP/1.1" 301 206 "Mozilla/5.0"

10.0.0.5 - - [05/Nov/2026:10:22:10 +0530] "GET /index.html HTTP/1.1" 200 8655 "Googlebot/2.1"

192.168.1.22 - - [05/Nov/2026:10:22:55 +0530] "GET /api/users HTTP/1.1" 200 7909 "Googlebot/2.1"

10.0.0.8 - - [05/Nov/2026:10:23:25 +0530] "GET /api/users HTTP/1.1" 200 2881 "curl/8.4.0"

192.168.1.10 - - [05/Nov/2026:10:24:29 +0530] "POST /api/login HTTP/1.1" 401 234 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:24:52 +0530] "GET /admin HTTP/1.1" 403 266 "Mozilla/5.0"

172.16.0.3 - - [05/Nov/2026:10:25:57 +0530] "GET /favicon.ico HTTP/1.1" 404 284 "curl/8.4.0"

10.0.0.5 - - [05/Nov/2026:10:26:40 +0530] "POST /api/login HTTP/1.1" 401 258 "Mozilla/5.0"

192.168.1.22 - - [05/Nov/2026:10:28:06 +0530] "GET /contact.html HTTP/1.1" 200 13649 "Mozilla/5.0"

192.168.1.22 - - [05/Nov/2026:10:28:38 +0530] "GET /index.html HTTP/1.1" 200 13251 "Mozilla/5.0"

192.168.1.22 - - [05/Nov/2026:10:29:33 +0530] "GET /contact.html HTTP/1.1" 200 9119 "curl/8.4.0"

192.168.1.15 - - [05/Nov/2026:10:30:08 +0530] "GET /Admin/settings HTTP/1.1" 403 218 "Mozilla/5.0"

203.0.113.45 - - [05/Nov/2026:10:30:47 +0530] "GET /old-page HTTP/1.1" 301 185 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:31:35 +0530] "GET /api/orders HTTP/1.1" 500 463 "Googlebot/2.1"

192.168.1.10 - - [05/Nov/2026:10:32:41 +0530] "POST /api/orders HTTP/1.1" 503 451 "python-requests/2.31"

192.168.1.10 - - [05/Nov/2026:10:33:09 +0530] "PUT /api/users/12 HTTP/1.1" 200 2516 "Mozilla/5.0"

172.16.0.3 - - [05/Nov/2026:10:34:23 +0530] "DELETE /api/users/7 HTTP/1.1" 204 0 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:35:18 +0530] "GET /api/users HTTP/1.1" 200 7768 "python-requests/2.31"

192.168.1.10 - - [05/Nov/2026:10:36:06 +0530] "POST /api/login HTTP/1.1" 200 8234 "Mozilla/5.0"

10.0.0.5 - - [05/Nov/2026:10:37:33 +0530] "POST /api/login HTTP/1.1" 200 3445 "python-requests/2.31"

192.168.1.15 - - [05/Nov/2026:10:38:19 +0530] "GET /favicon.ico HTTP/1.1" 404 214 "Mozilla/5.0"

10.0.0.5 - - [05/Nov/2026:10:39:43 +0530] "POST /api/users HTTP/1.1" 201 443 "python-requests/2.31"

203.0.113.45 - - [05/Nov/2026:10:40:53 +0530] "GET /admin HTTP/1.1" 403 250 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:41:29 +0530] "DELETE /api/users/7 HTTP/1.1" 204 0 "Mozilla/5.0"

10.0.0.5 - - [05/Nov/2026:10:42:20 +0530] "GET /about.html HTTP/1.1" 200 8327 "Mozilla/5.0"

192.168.1.15 - - [05/Nov/2026:10:43:50 +0530] "GET /about.html HTTP/1.1" 200 5871 "curl/8.4.0"

192.168.1.22 - - [05/Nov/2026:10:44:45 +0530] "POST /api/users HTTP/1.1" 201 458 "python-requests/2.31"

192.168.1.15 - - [05/Nov/2026:10:45:10 +0530] "GET /admin HTTP/1.1" 403 349 "Googlebot/2.1"

10.0.0.8 - - [05/Nov/2026:10:46:19 +0530] "GET /contact.html HTTP/1.1" 200 3248 "Googlebot/2.1"

Q10. Print only the IP addresses that start with 10. (print just the matching part, not the whole line).

Command: grep -Eo '^10\.[0-9]+\.[0-9]+\.[0-9]+' access.log

Output:

10.0.0.5

10.0.0.8

10.0.0.5

10.0.0.5

10.0.0.8

10.0.0.5

10.0.0.8

10.0.0.5

10.0.0.5

10.0.0.5

10.0.0.5

10.0.0.8

Section B: cut (Questions 11 to 20)

Q11. Print only the IP address from every line.

Command: cut -d' ' -f1 access.log

Output:

203.0.113.45

192.168.1.10

10.0.0.5

192.168.1.10

192.168.1.15

192.168.1.15

192.168.1.15

10.0.0.8

192.168.1.22

192.168.1.10

192.168.1.15

10.0.0.5

192.168.1.10

203.0.113.45

192.168.1.10

192.168.1.10

10.0.0.5

192.168.1.10

192.168.1.15

10.0.0.8

192.168.1.15

192.168.1.22

10.0.0.5

192.168.1.22

10.0.0.8

192.168.1.10

192.168.1.15

172.16.0.3

10.0.0.5

192.168.1.22

192.168.1.22

192.168.1.22

192.168.1.15

203.0.113.45

192.168.1.10

192.168.1.10

192.168.1.10

172.16.0.3

192.168.1.10

192.168.1.10

10.0.0.5

192.168.1.15

10.0.0.5

203.0.113.45

192.168.1.10

10.0.0.5

192.168.1.15

192.168.1.22

192.168.1.15

10.0.0.8

Q12. Print the IP address and the status code of every request.

Command: cut -d' ' -f1,9 access.log

Output:

203.0.113.45 404

192.168.1.10 500

10.0.0.5 200

192.168.1.10 200

192.168.1.15 200

192.168.1.15 200

192.168.1.15 200

10.0.0.8 403

192.168.1.22 200

192.168.1.10 503

192.168.1.15 200

10.0.0.5 301

192.168.1.10 200

203.0.113.45 404

192.168.1.10 404

192.168.1.10 401

10.0.0.5 301

192.168.1.10 200

192.168.1.15 200

10.0.0.8 404

192.168.1.15 200

192.168.1.22 301

10.0.0.5 200

192.168.1.22 200

10.0.0.8 200

192.168.1.10 401

192.168.1.15 403

172.16.0.3 404

10.0.0.5 401

192.168.1.22 200

192.168.1.22 200

192.168.1.22 200

192.168.1.15 403

203.0.113.45 301

192.168.1.10 500

192.168.1.10 503

192.168.1.10 200

172.16.0.3 204

192.168.1.10 200

192.168.1.10 200

10.0.0.5 200

192.168.1.15 404

10.0.0.5 201

203.0.113.45 403

192.168.1.10 204

10.0.0.5 200

192.168.1.15 200

192.168.1.22 201

192.168.1.15 403

10.0.0.8 200

Q13. Print only the request part that sits inside the first pair of double quotes, for example `GET /index.html HTTP/1.1`.

Command: cut -d'"' -f2 access.log

Output:

GET /missing.html HTTP/1.1

GET /api/orders HTTP/1.1

GET /about.html HTTP/1.1

POST /api/login HTTP/1.1

GET /images/logo.png HTTP/1.1

PUT /api/users/12 HTTP/1.1

GET /images/logo.png HTTP/1.1

GET /Admin/settings HTTP/1.1

GET /images/logo.png HTTP/1.1

POST /api/orders HTTP/1.1

GET /contact.html HTTP/1.1

GET /old-page HTTP/1.1

GET /api/users HTTP/1.1

GET /favicon.ico HTTP/1.1

GET /missing.html HTTP/1.1

POST /api/login HTTP/1.1

GET /old-page HTTP/1.1

POST /api/login HTTP/1.1

GET /index.html HTTP/1.1

GET /favicon.ico HTTP/1.1

GET /contact.html HTTP/1.1

GET /old-page HTTP/1.1

GET /index.html HTTP/1.1

GET /api/users HTTP/1.1

GET /api/users HTTP/1.1

POST /api/login HTTP/1.1

GET /admin HTTP/1.1

GET /favicon.ico HTTP/1.1

POST /api/login HTTP/1.1

GET /contact.html HTTP/1.1

GET /index.html HTTP/1.1

GET /contact.html HTTP/1.1

GET /Admin/settings HTTP/1.1

GET /old-page HTTP/1.1

GET /api/orders HTTP/1.1

POST /api/orders HTTP/1.1

PUT /api/users/12 HTTP/1.1

DELETE /api/users/7 HTTP/1.1

GET /api/users HTTP/1.1

POST /api/login HTTP/1.1

POST /api/login HTTP/1.1

GET /favicon.ico HTTP/1.1

POST /api/users HTTP/1.1

GET /admin HTTP/1.1

DELETE /api/users/7 HTTP/1.1

GET /about.html HTTP/1.1

GET /about.html HTTP/1.1

POST /api/users HTTP/1.1

GET /admin HTTP/1.1

GET /contact.html HTTP/1.1

Q14. Print the full timestamp without the square brackets, for example `05/Nov/2026:10:01:01 +0530`.

Command: cut -d'[' -f2 access.log | cut -d']' -f1

Output:

05/Nov/2026:10:01:01 +0530

05/Nov/2026:10:01:48 +0530

05/Nov/2026:10:02:36 +0530

05/Nov/2026:10:03:49 +0530

05/Nov/2026:10:04:33 +0530

05/Nov/2026:10:05:51 +0530

05/Nov/2026:10:07:08 +0530

05/Nov/2026:10:07:37 +0530

05/Nov/2026:10:08:31 +0530

05/Nov/2026:10:09:35 +0530

05/Nov/2026:10:10:26 +0530

05/Nov/2026:10:11:21 +0530

05/Nov/2026:10:12:10 +0530

05/Nov/2026:10:13:10 +0530

05/Nov/2026:10:14:20 +0530

05/Nov/2026:10:14:46 +0530

05/Nov/2026:10:15:25 +0530

05/Nov/2026:10:16:24 +0530

05/Nov/2026:10:17:51 +0530

05/Nov/2026:10:18:56 +0530

05/Nov/2026:10:19:45 +0530

05/Nov/2026:10:20:49 +0530

05/Nov/2026:10:22:10 +0530

05/Nov/2026:10:22:55 +0530

05/Nov/2026:10:23:25 +0530

05/Nov/2026:10:24:29 +0530

05/Nov/2026:10:24:52 +0530

05/Nov/2026:10:25:57 +0530

05/Nov/2026:10:26:40 +0530

05/Nov/2026:10:28:06 +0530

05/Nov/2026:10:28:38 +0530

05/Nov/2026:10:29:33 +0530

05/Nov/2026:10:30:08 +0530

05/Nov/2026:10:30:47 +0530

05/Nov/2026:10:31:35 +0530

05/Nov/2026:10:32:41 +0530

05/Nov/2026:10:33:09 +0530

05/Nov/2026:10:34:23 +0530

05/Nov/2026:10:35:18 +0530

05/Nov/2026:10:36:06 +0530

05/Nov/2026:10:37:33 +0530

05/Nov/2026:10:38:19 +0530

05/Nov/2026:10:39:43 +0530

05/Nov/2026:10:40:53 +0530

05/Nov/2026:10:41:29 +0530

05/Nov/2026:10:42:20 +0530

05/Nov/2026:10:43:50 +0530

05/Nov/2026:10:44:45 +0530

05/Nov/2026:10:45:10 +0530

05/Nov/2026:10:46:19 +0530

Q15. Print only the time of each request in HH:MM:SS format, for example `10:01:01`.

Command: cut -d'[' -f2 access.log | cut -d']' -f1 | cut -d' ' -f1 | cut -d':' -f2-4

Output:

10:01:01

10:01:48

10:02:36

10:03:49

10:04:33

10:05:51

10:07:08

10:07:37

10:08:31

10:09:35

10:10:26

10:11:21

10:12:10

10:13:10

10:14:20

10:14:46

10:15:25

10:16:24

10:17:51

10:18:56

10:19:45

10:20:49

10:22:10

10:22:55

10:23:25

10:24:29

10:24:52

10:25:57

10:26:40

10:28:06

10:28:38

10:29:33

10:30:08

10:30:47

10:31:35

10:32:41

10:33:09

10:34:23

10:35:18

10:36:06

10:37:33

10:38:19

10:39:43

10:40:53

10:41:29

10:42:20

10:43:50

10:44:45

10:45:10

10:46:19

Q16. Print only the HTTP method (GET, POST, PUT or DELETE) of each request.

Command:

Output:

Q17. Print the first 15 characters of every line.

Command:

Output:

Q18. Print the list of unique IP addresses, sorted.

Command:

Output:

Q19. Count how many requests came from each IP address, with the busiest IP at the top.

Command:

Output:

Q20. Print only the user agent of each request, for example `Mozilla/5.0`.

Command:

Output:

Section C: awk (Questions 21 to 30)

Q21. Print the IP address and the requested path of every line.

Command: awk '{print $1 $7}' access.log

Output:

203.0.113.45 /missing.html

192.168.1.10 /api/orders

10.0.0.5 /about.html

192.168.1.10 /api/login

192.168.1.15 /images/logo.png

192.168.1.15 /api/users/12

192.168.1.15 /images/logo.png

10.0.0.8 /Admin/settings

192.168.1.22 /images/logo.png

192.168.1.10 /api/orders

192.168.1.15 /contact.html

10.0.0.5 /old-page

192.168.1.10 /api/users

203.0.113.45 /favicon.ico

192.168.1.10 /missing.html

192.168.1.10 /api/login

10.0.0.5 /old-page

192.168.1.10 /api/login

192.168.1.15 /index.html

10.0.0.8 /favicon.ico

192.168.1.15 /contact.html

192.168.1.22 /old-page

10.0.0.5 /index.html

192.168.1.22 /api/users

10.0.0.8 /api/users

192.168.1.10 /api/login

192.168.1.15 /admin

172.16.0.3 /favicon.ico

10.0.0.5 /api/login

192.168.1.22 /contact.html

192.168.1.22 /index.html

192.168.1.22 /contact.html

192.168.1.15 /Admin/settings

203.0.113.45 /old-page

192.168.1.10 /api/orders

192.168.1.10 /api/orders

192.168.1.10 /api/users/12

172.16.0.3 /api/users/7

192.168.1.10 /api/users

192.168.1.10 /api/login

10.0.0.5 /api/login

192.168.1.15 /favicon.ico

10.0.0.5 /api/users

203.0.113.45 /admin

192.168.1.10 /api/users/7

10.0.0.5 /about.html

192.168.1.15 /about.html

192.168.1.22 /api/users

192.168.1.15 /admin

10.0.0.8 /contact.html

Q22. Print only the lines where the status code is exactly 500.

Command: awk '$9 == 500' access.log

Output:

192.168.1.10 - - [05/Nov/2026:10:01:48 +0530] "GET /api/orders HTTP/1.1" 500 330 "Mozilla/5.0"

192.168.1.10 - - [05/Nov/2026:10:31:35 +0530] "GET /api/orders HTTP/1.1" 500 463 "Googlebot/2.1"

Q23. Calculate the total number of bytes sent across all requests.

Command: awk '{total+=$10} END {print total}' access.log

Output:

239481

Q24. Count how many requests there are for each status code.

Command: awk '{count[$9]++} END {for(i in count) print i, count[i]}' access.log

Output:

200 24

201 2

204 2

301 4

401 3

403 5

404 6

500 2

503 2

Q25. Print the IP, path and size of every response larger than 10000 bytes.

Command: awk ' $10>10000 {print 1,7,$10}' access.log

Output:

10.0.0.5 /about.html 10255

192.168.1.10 /api/login 10153

192.168.1.15 /images/logo.png 20573

192.168.1.15 /api/users/12 13815

192.168.1.15 /images/logo.png 21454

192.168.1.22 /images/logo.png 21160

192.168.1.10 /api/login 13082

192.168.1.15 /index.html 13221

192.168.1.22 /contact.html 13649

192.168.1.22 /index.html 13251

Q26. Calculate the average response size in bytes.

Command: awk '{total+=$10; count++} END {print total/count}' access.log

Output:

4789.62

Q27. Print each line with its line number in front, then print the total number of lines at the end.

Command: awk '{print NR, $0} END {print "total lines:",NR}' access.log

Output:

1 203.0.113.45 - - [05/Nov/2026:10:01:01 +0530] "GET /missing.html HTTP/1.1" 404 243 "python-requests/2.31"

2 192.168.1.10 - - [05/Nov/2026:10:01:48 +0530] "GET /api/orders HTTP/1.1" 500 330 "Mozilla/5.0"

3 10.0.0.5 - - [05/Nov/2026:10:02:36 +0530] "GET /about.html HTTP/1.1" 200 10255 "Mozilla/5.0"

4 192.168.1.10 - - [05/Nov/2026:10:03:49 +0530] "POST /api/login HTTP/1.1" 200 10153 "Googlebot/2.1"

5 192.168.1.15 - - [05/Nov/2026:10:04:33 +0530] "GET /images/logo.png HTTP/1.1" 200 20573 "curl/8.4.0"

6 192.168.1.15 - - [05/Nov/2026:10:05:51 +0530] "PUT /api/users/12 HTTP/1.1" 200 13815 "Mozilla/5.0"

7 192.168.1.15 - - [05/Nov/2026:10:07:08 +0530] "GET /images/logo.png HTTP/1.1" 200 21454 "Mozilla/5.0"

8 10.0.0.8 - - [05/Nov/2026:10:07:37 +0530] "GET /Admin/settings HTTP/1.1" 403 327 "Mozilla/5.0"

9 192.168.1.22 - - [05/Nov/2026:10:08:31 +0530] "GET /images/logo.png HTTP/1.1" 200 21160 "Googlebot/2.1"

10 192.168.1.10 - - [05/Nov/2026:10:09:35 +0530] "POST /api/orders HTTP/1.1" 503 447 "Mozilla/5.0"

11 192.168.1.15 - - [05/Nov/2026:10:10:26 +0530] "GET /contact.html HTTP/1.1" 200 2120 "python-requests/2.31"

12 10.0.0.5 - - [05/Nov/2026:10:11:21 +0530] "GET /old-page HTTP/1.1" 301 209 "Googlebot/2.1"

13 192.168.1.10 - - [05/Nov/2026:10:12:10 +0530] "GET /api/users HTTP/1.1" 200 5104 "python-requests/2.31"

14 203.0.113.45 - - [05/Nov/2026:10:13:10 +0530] "GET /favicon.ico HTTP/1.1" 404 293 "curl/8.4.0"

15 192.168.1.10 - - [05/Nov/2026:10:14:20 +0530] "GET /missing.html HTTP/1.1" 404 191 "python-requests/2.31"

16 192.168.1.10 - - [05/Nov/2026:10:14:46 +0530] "POST /api/login HTTP/1.1" 401 257 "curl/8.4.0"

17 10.0.0.5 - - [05/Nov/2026:10:15:25 +0530] "GET /old-page HTTP/1.1" 301 271 "curl/8.4.0"

18 192.168.1.10 - - [05/Nov/2026:10:16:24 +0530] "POST /api/login HTTP/1.1" 200 13082 "Mozilla/5.0"

19 192.168.1.15 - - [05/Nov/2026:10:17:51 +0530] "GET /index.html HTTP/1.1" 200 13221 "Mozilla/5.0"

20 10.0.0.8 - - [05/Nov/2026:10:18:56 +0530] "GET /favicon.ico HTTP/1.1" 404 199 "Googlebot/2.1"

21 192.168.1.15 - - [05/Nov/2026:10:19:45 +0530] "GET /contact.html HTTP/1.1" 200 6625 "Googlebot/2.1"

22 192.168.1.22 - - [05/Nov/2026:10:20:49 +0530] "GET /old-page HTTP/1.1" 301 206 "Mozilla/5.0"

23 10.0.0.5 - - [05/Nov/2026:10:22:10 +0530] "GET /index.html HTTP/1.1" 200 8655 "Googlebot/2.1"

24 192.168.1.22 - - [05/Nov/2026:10:22:55 +0530] "GET /api/users HTTP/1.1" 200 7909 "Googlebot/2.1"

25 10.0.0.8 - - [05/Nov/2026:10:23:25 +0530] "GET /api/users HTTP/1.1" 200 2881 "curl/8.4.0"

26 192.168.1.10 - - [05/Nov/2026:10:24:29 +0530] "POST /api/login HTTP/1.1" 401 234 "Mozilla/5.0"

27 192.168.1.15 - - [05/Nov/2026:10:24:52 +0530] "GET /admin HTTP/1.1" 403 266 "Mozilla/5.0"

28 172.16.0.3 - - [05/Nov/2026:10:25:57 +0530] "GET /favicon.ico HTTP/1.1" 404 284 "curl/8.4.0"

29 10.0.0.5 - - [05/Nov/2026:10:26:40 +0530] "POST /api/login HTTP/1.1" 401 258 "Mozilla/5.0"

30 192.168.1.22 - - [05/Nov/2026:10:28:06 +0530] "GET /contact.html HTTP/1.1" 200 13649 "Mozilla/5.0"

31 192.168.1.22 - - [05/Nov/2026:10:28:38 +0530] "GET /index.html HTTP/1.1" 200 13251 "Mozilla/5.0"

32 192.168.1.22 - - [05/Nov/2026:10:29:33 +0530] "GET /contact.html HTTP/1.1" 200 9119 "curl/8.4.0"

33 192.168.1.15 - - [05/Nov/2026:10:30:08 +0530] "GET /Admin/settings HTTP/1.1" 403 218 "Mozilla/5.0"

34 203.0.113.45 - - [05/Nov/2026:10:30:47 +0530] "GET /old-page HTTP/1.1" 301 185 "Mozilla/5.0"

35 192.168.1.10 - - [05/Nov/2026:10:31:35 +0530] "GET /api/orders HTTP/1.1" 500 463 "Googlebot/2.1"

36 192.168.1.10 - - [05/Nov/2026:10:32:41 +0530] "POST /api/orders HTTP/1.1" 503 451 "python-requests/2.31"

37 192.168.1.10 - - [05/Nov/2026:10:33:09 +0530] "PUT /api/users/12 HTTP/1.1" 200 2516 "Mozilla/5.0"

38 172.16.0.3 - - [05/Nov/2026:10:34:23 +0530] "DELETE /api/users/7 HTTP/1.1" 204 0 "Mozilla/5.0"

39 192.168.1.10 - - [05/Nov/2026:10:35:18 +0530] "GET /api/users HTTP/1.1" 200 7768 "python-requests/2.31"

40 192.168.1.10 - - [05/Nov/2026:10:36:06 +0530] "POST /api/login HTTP/1.1" 200 8234 "Mozilla/5.0"

41 10.0.0.5 - - [05/Nov/2026:10:37:33 +0530] "POST /api/login HTTP/1.1" 200 3445 "python-requests/2.31"

42 192.168.1.15 - - [05/Nov/2026:10:38:19 +0530] "GET /favicon.ico HTTP/1.1" 404 214 "Mozilla/5.0"

43 10.0.0.5 - - [05/Nov/2026:10:39:43 +0530] "POST /api/users HTTP/1.1" 201 443 "python-requests/2.31"

44 203.0.113.45 - - [05/Nov/2026:10:40:53 +0530] "GET /admin HTTP/1.1" 403 250 "Mozilla/5.0"

45 192.168.1.10 - - [05/Nov/2026:10:41:29 +0530] "DELETE /api/users/7 HTTP/1.1" 204 0 "Mozilla/5.0"

46 10.0.0.5 - - [05/Nov/2026:10:42:20 +0530] "GET /about.html HTTP/1.1" 200 8327 "Mozilla/5.0"

47 192.168.1.15 - - [05/Nov/2026:10:43:50 +0530] "GET /about.html HTTP/1.1" 200 5871 "curl/8.4.0"

48 192.168.1.22 - - [05/Nov/2026:10:44:45 +0530] "POST /api/users HTTP/1.1" 201 458 "python-requests/2.31"

49 192.168.1.15 - - [05/Nov/2026:10:45:10 +0530] "GET /admin HTTP/1.1" 403 349 "Googlebot/2.1"

50 10.0.0.8 - - [05/Nov/2026:10:46:19 +0530] "GET /contact.html HTTP/1.1" 200 3248 "Googlebot/2.1"

total lines: 50

Q28. For POST requests only, print the path and the status code.

Command: awk '$6 ~ /POST/ {print $7, $9}' access.log

Output:

/api/login 200

/api/orders 503

/api/login 401

/api/login 200

/api/login 401

/api/login 401

/api/orders 503

/api/login 200

/api/login 200

/api/users 201

/api/users 201

Q29. Find the single IP address that made the most requests, and print it with its count.

Command: awk '{print $1}' access.log | sort | uniq -c |sort -nr | head -1

Output:

14 192.168.1.10

Q30. Use printf to print a neat report line for every request, in this format: `IP: 10.0.0.5 | Status: 200 | Size: 10.01 KB` (size in KB with 2 decimals).

Command: awk '{printf "IP: %s | Status: %s | Size: %.2f KB\n",1,9,$10/1024}' access.log

Output:

IP: 203.0.113.45 | Status: 404 | Size: 0.24 KB

IP: 192.168.1.10 | Status: 500 | Size: 0.32 KB

IP: 10.0.0.5 | Status: 200 | Size: 10.01 KB

IP: 192.168.1.10 | Status: 200 | Size: 9.92 KB

IP: 192.168.1.15 | Status: 200 | Size: 20.09 KB

IP: 192.168.1.15 | Status: 200 | Size: 13.49 KB

IP: 192.168.1.15 | Status: 200 | Size: 20.95 KB

IP: 10.0.0.8 | Status: 403 | Size: 0.32 KB

IP: 192.168.1.22 | Status: 200 | Size: 20.66 KB

IP: 192.168.1.10 | Status: 503 | Size: 0.44 KB

IP: 192.168.1.15 | Status: 200 | Size: 2.07 KB

IP: 10.0.0.5 | Status: 301 | Size: 0.20 KB

IP: 192.168.1.10 | Status: 200 | Size: 4.98 KB

IP: 203.0.113.45 | Status: 404 | Size: 0.29 KB

IP: 192.168.1.10 | Status: 404 | Size: 0.19 KB

IP: 192.168.1.10 | Status: 401 | Size: 0.25 KB

IP: 10.0.0.5 | Status: 301 | Size: 0.26 KB

IP: 192.168.1.10 | Status: 200 | Size: 12.78 KB

IP: 192.168.1.15 | Status: 200 | Size: 12.91 KB

IP: 10.0.0.8 | Status: 404 | Size: 0.19 KB

IP: 192.168.1.15 | Status: 200 | Size: 6.47 KB

IP: 192.168.1.22 | Status: 301 | Size: 0.20 KB

IP: 10.0.0.5 | Status: 200 | Size: 8.45 KB

IP: 192.168.1.22 | Status: 200 | Size: 7.72 KB

IP: 10.0.0.8 | Status: 200 | Size: 2.81 KB

IP: 192.168.1.10 | Status: 401 | Size: 0.23 KB

IP: 192.168.1.15 | Status: 403 | Size: 0.26 KB

IP: 172.16.0.3 | Status: 404 | Size: 0.28 KB

IP: 10.0.0.5 | Status: 401 | Size: 0.25 KB

IP: 192.168.1.22 | Status: 200 | Size: 13.33 KB

IP: 192.168.1.22 | Status: 200 | Size: 12.94 KB

IP: 192.168.1.22 | Status: 200 | Size: 8.91 KB

IP: 192.168.1.15 | Status: 403 | Size: 0.21 KB

IP: 203.0.113.45 | Status: 301 | Size: 0.18 KB

IP: 192.168.1.10 | Status: 500 | Size: 0.45 KB

IP: 192.168.1.10 | Status: 503 | Size: 0.44 KB

IP: 192.168.1.10 | Status: 200 | Size: 2.46 KB

IP: 172.16.0.3 | Status: 204 | Size: 0.00 KB

IP: 192.168.1.10 | Status: 200 | Size: 7.59 KB

IP: 192.168.1.10 | Status: 200 | Size: 8.04 KB

IP: 10.0.0.5 | Status: 200 | Size: 3.36 KB

IP: 192.168.1.15 | Status: 404 | Size: 0.21 KB

IP: 10.0.0.5 | Status: 201 | Size: 0.43 KB

IP: 203.0.113.45 | Status: 403 | Size: 0.24 KB

IP: 192.168.1.10 | Status: 204 | Size: 0.00 KB

IP: 10.0.0.5 | Status: 200 | Size: 8.13 KB

IP: 192.168.1.15 | Status: 200 | Size: 5.73 KB

IP: 192.168.1.22 | Status: 201 | Size: 0.45 KB

IP: 192.168.1.15 | Status: 403 | Size: 0.34 KB

IP: 10.0.0.8 | Status: 200 | Size: 3.17 KB

Bonus (optional)

Using one awk command, print the number of failed login attempts (POST /api/login with status 401) for each IP address.